

$\gamma\delta$ T cells at the interface of innate and adaptive immunity in cancer

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ABSTRACT

$\gamma\delta$ T cells are unconventional lymphocytes that bridge innate and adaptive immunity by combining recognition of stress-induced ligands independently of classical major histocompatibility complex molecules with the capacity to undergo clonal expansion and long-term adaptation. Their unusual ability to detect malignant transformation using semi-invariant T-cell receptors, butyrophilin recognition and natural killer-like receptors positions them as powerful effector cells in tumors that evade classical immune escape mechanisms. Furthermore, distinct $\gamma\delta$ subsets have distinct phenotyping and specific tissue-residencies, which could be leveraged to modulate immunological responses. We evaluate engineered therapies and different experimental platforms for studying $\gamma\delta$ T cell biology. We conclude that next-generation cancer treatments should strategically integrate $\gamma\delta$ T cells into synthetic immunology, individualized modeling, and combinatorial regimes.

INTRODUCTION $\gamma\delta$ T-CELL OVERVIEW

The concept of innate versus adaptive immunity was developed during the first half of the 20th century and has guided immunological research for multiple decades.¹ However, recent discoveries of new immune populations have challenged this binary classification, like $\gamma\delta$ T cells.¹ These immune cells have demonstrated a dual nature by using innate pattern recognition alongside rapid effector functions and adaptive features including antigen specificity, clonal expansion, and memory-like phenotype.²

Unlike $\alpha\beta$ T cells, $\gamma\delta$ T cells detect various ligands independently of classical major histocompatibility complex (MHC) molecules, including phosphoantigens, lipids, and non-classical stress-induced molecules.^{3,4} Their semi-invariant T-cell receptors (TCRs), natural killer (NK)-like receptors, and pattern recognition receptors (PRRs) enable rapid detection of cellular stress and malignant transformations, outside conventional antigen-processing pathways.²

The $\gamma\delta$ T-cell population consists of a heterogeneous lymphocyte lineage shaped by TCR repertoire constraints, subset-specific transcriptional programs,⁵ and tissue-derived

signals.^{5–7} Their subset classification, distinct tissue tropism, and dominant functional programs vary across species¹ (table 1).

Regardless of species-specific differences, core functional hallmarks of $\gamma\delta$ T cells are conserved.⁸ For example, the capacity of rapid cytokine expression, killing activity, stress sensing, as well as the ability to shape other immune effectors in the TME.⁹ Indeed, these attributes render $\gamma\delta$ T cells uniquely capable of mediating the bridge between innate and adaptive immunity and make them a potentially valuable tool in the arsenal against cancer.²

This review focuses on subset specialization, tissue localization, effector programming and related biology across mice and humans to provide a comparative and translational perspective on $\gamma\delta$ T cells in cancer. We also examine experimental platforms and therapeutic models, highlighting how $\gamma\delta$ T-cell biology can overcome limitations of existing $\alpha\beta$ T cell-targeted therapies, including bispecific antibodies and CAR-T cells, while summarizing barriers to clinical translation.

$\gamma\delta$ T-CELL DEVELOPMENT AND SUBSET DIVERSITY

Functional heterogeneity and immune functions of $\gamma\delta$ T cells are shaped by their developmental and subset-restricted differentiation, which differs prominently between humans and mice.^{8,10} Somatic V(D)J recombination of TRGV and TRDV loci is strictly temporally and anatomically controlled, generating different $\gamma\delta$ subsets with specified TCR usage, transcription programs, and tissue¹¹ residency.^{9,11,12} Although species-specific differences in development and repertoire organization, conserved roles in stress surveillance, cytokine-driven immunity, barrier maintenance, and tumor-associated immune responses are preserved across species.^{12,13} Hence, understanding

Table 1 Functional and phenotypic comparison of major $\gamma\delta$ T-cell subsets in mice and humans

Subset	Primary tissue tropism	TCR usage	Core markers and transcription factors	Cytokine potential	Notable functions	Ref
Mouse						
$V\gamma 5^+$ (DETCs)	Epidermis	$V\gamma 5-V\delta 1$ (restricted)	Egr3, Sox13, Runx3; CD27 ⁺ ; CD122 ⁺ (variable); CD103 ⁺ ; CD45RB ^{hi}	IFN- γ , insulin-like growth factor (IGF-1), keratinocyte growth factor (KGF)	Epidermal stress surveillance, wound repair, maintenance of barrier integrity; tumor roles depend on epithelial damage context.	18
$V\gamma 6^+$	Mucosal and epithelial tissues (lung, uterus, oral mucosa, gut)	$V\gamma 6-V\delta 1/V\delta 6$	CCR6 ⁺ , CD44 ⁺ , CD27 ⁺ , IL-23R ⁺ ; ROR γ t, SOX13, Blk Tissue resident phenotype CD69 ⁺ CD103 ⁺ PD-1 ⁺	IL-17A	Cytokine-driven innate IL-17A responses; early antimicrobial defense; context-dependent protumor or antitumor functions.	20
$V\gamma 4^+$	Peripheral and inflamed tissues	$V\gamma 4-V\delta 4/V\delta 5$	CD44 ^{hi} CD27 ⁺ CCR6 ⁺ IL-23R ⁺ - ROR γ t, SOX13, Blk (IL-17A bias). CD27 ⁺ CD62L ⁺ T-bet ⁺ (IFN- γ bias)	IL-17A; IFN- γ (plastic)	Highly plastic subset bridging IL-17A-driven inflammation and type-1 immunity; antitumor activity under IL-12/type-1 conditions.	1 25 98
$V\gamma 7^+$	Intestinal epithelium, mesenteric lymph nodes	$V\gamma 7-V\delta 4$ /others	CD8 $\alpha\alpha$ ⁺ , CD103 ⁺ , CD69 ⁺ , CCR9 ⁺ , CD122 ⁺ ; T-bet, Eomes	IFN- γ	IEL-mediated epithelial homeostasis and microbial defense; limited evidence in cancer	10
$V\gamma 1^+$	Lymphoid and systemic compartments	$V\gamma 1-V\delta 2/V\delta 5$	CD27 ⁺ CD45RB ⁺ CCR6 ⁺ ; T-bet, Eomes	IFN- γ , perforin, granzymes	Potent cytotoxic and antiviral functions; consistent antitumor activity; murine analog of human $V\gamma 9V\delta 2^+$ effector programs	2
Human						
$V\delta 2^+$	Circulating blood (systemic)	$V\delta 2+V\gamma 9/V\gamma X$	CD161 ⁺ , NKG2A ⁺ , CCR5 ⁺ ; CD16 ⁺ after activation; T-bet, Eomes; CD27 ⁺ CD45RA ⁺ naïve-like CD27 ⁺ CD45RA ⁺ or CD27 ⁺ CD45RA ⁻ effector memory-like states BTN3A1/BTN2A1-responsive	IFN- γ , TNF- α , granzyme B, perforin	Major phosphoantigen-responsive subset; broad stress-ligand cytotoxicity; principal target for $\gamma\delta$ -based cancer immunotherapy.	27 28
$V\delta 1^+$	Epithelial and mucosal tissues (skin, gut, lung, liver)	$V\delta 1$ -diverse $V\gamma$	CD103 ⁺ , CD69 ⁺ , CD38 ⁺ , CD101 ⁺ , CXCR6 ⁺ , TIGIT ⁺ , PLZF, ROR γ t, T-bet, Eomes	Cytotoxic mediators; IL-17A; IL-10; IFN- γ	Dominant tissue-resident $\gamma\delta$ subset; epithelial stress surveillance; antitumor activity in several tissues; protumor under IL-17A-skewed signals.	15
$V\delta 3^+$	Liver, biliary tract, mucosal tissues	$V\delta 3$ -varied $V\gamma$	NKG2D ⁺ , NKp30 ⁺ , CD56 ⁺ ; CD1-restricted lipid recognition	IFN- γ , TNF- α	Antiviral responses (eg, cytomegalovirus (CMV)) and hepatic immune surveillance; emerging antitumor properties.	35
$V\delta 5^+$	Mucosal surfaces, liver	$V\delta 5$ -varied $V\gamma$	Incompletely defined; often CD69 ⁺ CD103 ⁺ ; stress-responsive phenotype	IFN- γ , TNF- α (stress-induced)	Understudied subset; implicated in epithelial repair and mucosal homeostasis; potential roles in early tumor surveillance.	35 62
DETCs, dendritic epidermal T cells; IEL, intraepithelial lymphocyte; IFN, interferon; IL, interleukin; TCR, T-cell receptor; TNF, tumor necrosis factor.						

these similarities and differences between human and murine subsets regarding barrier defense and tumor immunity are crucial for accurately interpreting preclinical models and translating $\gamma\delta$ T-cell biology into clinical immunotherapy applications (figure 1).¹²

Subset specification and developmental programming in mice

Murine $\gamma\delta$ T cells develop in sequential thymic waves beginning during fetal life and continuing postnatally.¹⁴ Early fetal waves generate in a TdT-low environment, generating semi-invariant $V\gamma 5^+$, $V\gamma 6^+$ and $V\gamma 7^+$ repertoires that seed defined epithelial and mucosal

niches.^{12,13,15} In postnatal waves, increasing TdT expression permits the emergence of more diverse $V\gamma 1^+$ and $V\gamma 4^+$ repertoires, with broad junctional variability and heterogeneous $V\delta$ pairing.¹²

Effector fate is imprinted in the thymus and is determined by TCR signal strength, ontogeny, ligand availability, and cytokine cues.¹⁶ Strong TCR $\gamma\delta$ signals promote T-bet-driven interferon (IFN)- γ -biased lineages ($\gamma\delta^{\text{IFN-}\gamma}$), whereas weaker or absent signaling favors ROR γ t differentiation with stable IL-17A locus ($\gamma\delta^{17}$) accessibility.¹²

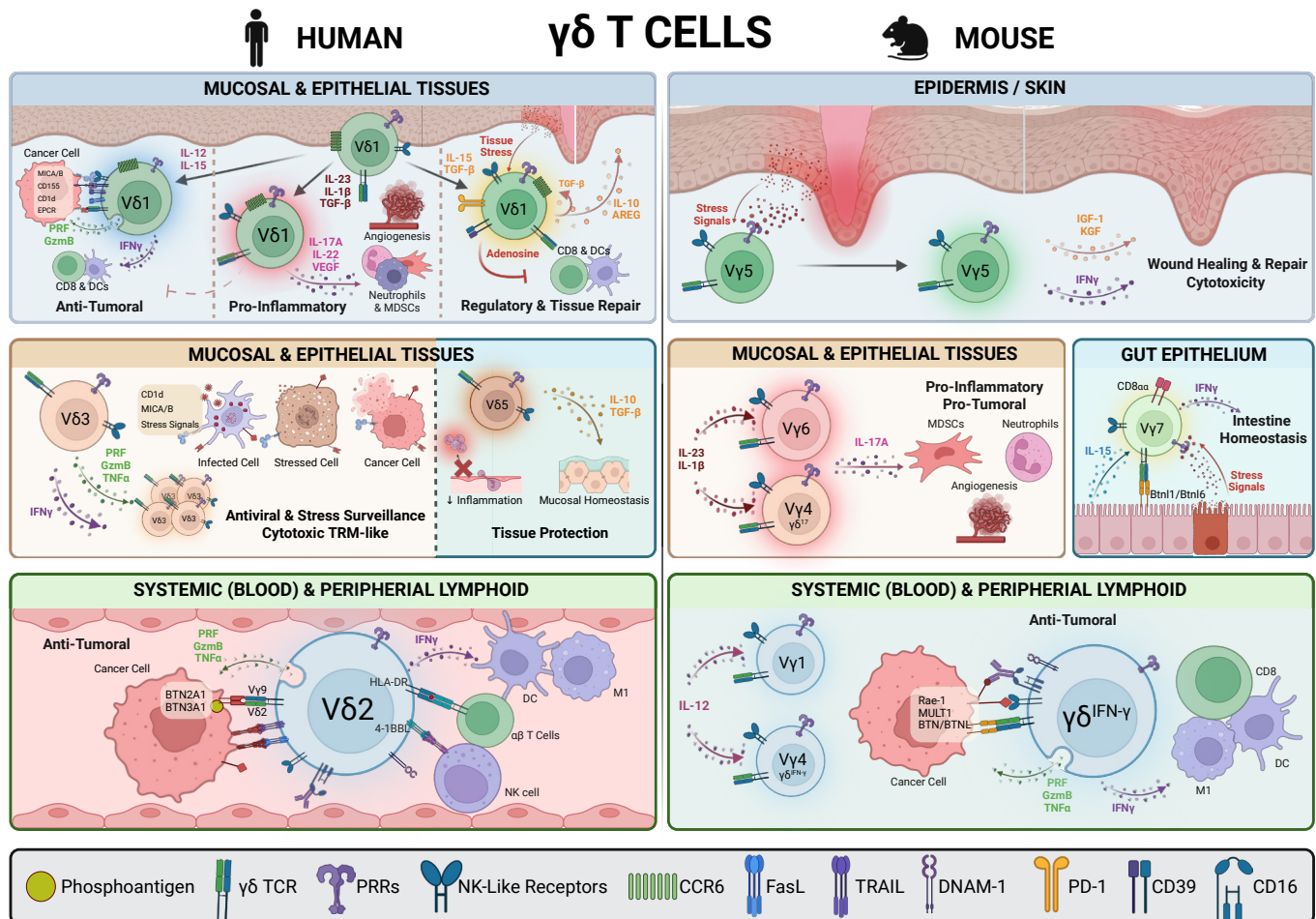


Figure 1 Comparative overview of $\gamma\delta$ T-cell subsets in humans and mice. On the left side, human $\gamma\delta$ T cells are highly specialized tissue and functionally plastic. Tissue-resident $V\delta 1^+$ cells integrate epithelial stress detection with three major effector programs: (1) type 1 cytotoxicity triggered by IL-12/IL-15 and T-bet/Eomes, defined by IFN- γ and perforin/granzyme release; (2) IL17A-mediated inflammation induced by IL23/IL1 β /TGF β associated with neutrophil and myeloid mobilization, as well as angiogenesis; and (3) regulatory and tissue repair programs supported by IL-15/TGF- β , including IL-10, amphiregulin, and CD39/CD73-dependent adenosine production. $V\delta 3^+$ cells contribute to antiviral, cancer and stress surveillance through lipid-mediated CD1d-restricted recognition and other ligands, $V\delta 5^+$ exhibit regulatory and tissue-protective properties associated with mucosal homeostasis. Peripheral $V\gamma 9V\delta 2^+$ T cells perceive phosphoantigens through the BTN3A1/BTN2A1 axis and display powerful cytotoxic activity, along with immune-regulatory activities including a temporary antigen-presenting-like capability. On the right, murine $\gamma\delta$ T cells are lineage-biased and tissue-restricted. $V\gamma 5^+$ dendritic epidermal T cells control skin homeostasis and wound healing through IFN- γ , IGF1 and KGF. $V\gamma 6^+$ and $V\gamma 4^+$ subsets of $\gamma\delta$ T cells mediate IL-17A-dependent barrier inflammation, while $V\gamma 4^+$ T cells retain plasticity to an IFN- γ -producing cytotoxic fate under IL-12. $V\gamma 7^+$ intraepithelial lymphocytes contribute to intestinal homeostasis by IFN- γ , whereas systemic $V\gamma 1^+$ cells exert potent antitumor responses through IFN- γ , perforin/granzyme-dependent cytotoxicity, and NK-like receptor-mediated mechanisms. Illustrations were generated with BioRender.com. IFN, interferon; IL, interleukin; NK, natural killer; PD-1, programmed cell death protein-1; TCR, T-cell receptor.

- ▶ $V\gamma 5^+$ dendritic epidermal T cells (DETCs): It is the first population to develop, produced during fetal development.^{8 17} They express a $V\gamma 5$ – $V\delta 1$ TCR, selected via Skint1 and form a long-lived epidermal network, mediating epithelial integrity.^{15 18}
 - ▶ $V\gamma 6^+$ cells: Shortly after DETCs, a wave of $V\gamma 6^+$ cells develops, characterized by $V\gamma 6$ – $V\delta 1$ TCR and IL-7/TGF- β -dependent $\gamma\delta^{17}$ program.¹⁹ After exiting the thymus, their phenotype promotes their recruitment and residency to mucosal and barrier tissues, where their cytokine-responsive programs contribute to rapid and early barrier immunity and inflammation.^{19–21}
 - ▶ $V\gamma 7^+$ cells: Emerging perinatally, these cells populate the gut epithelium, where they constitute a major $\gamma\delta$ intraepithelial lymphocyte (IEL) population.¹⁶ Their maintenance and long-term residency depend on local Btl1/Btl6-mediated TCR signals, and interleukin (IL)-15 supports a T-bet-driven type-1 cytotoxic profile, contributing to epithelial homeostasis.^{10 22 23}
 - ▶ $V\gamma 4^+$ cells: Postnatally generated, $V\gamma 4^+$ cells constitute a heterogeneous population which can adopt either $\gamma\delta^{17}$ or $\gamma\delta^{\text{IFN-}\gamma}$ effector fates.^{21 24} In peripheral sites, IL-12 can direct them toward IFN- γ -producing responses, while IL-1 β /IL-23 sustain $\gamma\delta^{17}$ phenotype, conferring pro-inflammatory or antitumor functions.^{20 24 25}
 - ▶ $V\gamma 1^+$ cells: Generated later postnatally, these cells exhibit the greatest junctional diversity among murine $V\gamma$ subsets giving rise to $\gamma\delta^{\text{IFN-}\gamma}$.²⁴ They are present in the systemic circulation and usually display a type-1 cytotoxic program that is functional analog of their human $V\gamma 9V\delta 2^+$ counterparts.^{1 8 10}
 - ▶ $V\delta 3^+$ T cells: These cells exhibit oligoclonal repertoires indicative of antigen-driven selection.⁹ They express NK and tissue-homing receptors consistent with hepatic and mucosal residence.³⁵ They can shift toward regulatory or repair-like states depending on the tissue environment.³⁶
 - ▶ $V\delta 5^+$ T cells: This rare and less-characterized subset is enriched in mucosal and hepatic tissues.³⁷ These cells are oligoclonal with evidence for antigen shaping.³⁵ Transcriptomic analysis shows a mixture of type-1 effector signatures combined with repair modules, suggesting internal heterogeneity and poorly understood developmental pathways.³⁵
- Phenotypic and functional features of murine and human subsets are summarized in [table 1](#).

BRIDGING INNATE AND ADAPTIVE IMMUNITY

$\gamma\delta$ T cells occupy a unique niche in the immune system at the crossroads of innate and adaptive traits.^{1 38} Contrary to conventional $\alpha\beta$ T lymphocytes, which need peptidic antigens presented by classical MHC molecules for their activation, several $\gamma\delta$ T-cell subsets respond to non-peptidic, or stress-induced ligands in an MHC-independent manner through semi-invariant $\gamma\delta$ TCRs and NK-like receptors and, in some cases, through PRRs.³⁸ This enables immediate recognition of dysregulated cells, especially at epithelial and mucosal surfaces.^{1 38 39} Despite this innate-like speed, subsets of $\gamma\delta$ T cells also exhibit adaptive hallmarks, including antigen-driven clonal expansion, long-term persistence, and functional plasticity, underscoring their hybrid immunological nature.³⁸

$\gamma\delta$ T-cell subset organization in humans

Human $\gamma\delta$ T cells are primarily organized by $V\delta$ usage and exhibit greater peripheral plasticity.¹ These cells acquire their effector functions in the periphery in response to cytokines, antigens, and inflammatory cues.⁹ Because the TCR δ locus is nested within the TCR α locus, $V\delta$ chains can pair with various $V\gamma$ elements, leading to a more versatile $\gamma\delta$ TCR repertoire with broad tissue distribution and diverse effector programming.^{3 11}

- ▶ $V\delta 2^+$ T cells: The major $\gamma\delta$ subset in adult peripheral blood, arising during fetal development and expanding postnatally in response to microbial and metabolic phosphoantigens.²⁶ This subset, when it pairs with $V\gamma 9$, senses phosphoantigens accumulation through BTN3A1/BTN2A1 complexes, where BTN3A1's intracellular B30.2 domain detects this accumulation, while BTN2A1 engages the germline-encoded region of $V\gamma 9$ to provide a canonical TCR interaction.^{27–29}
- ▶ $V\delta 1^+$ T cells: Resident in epithelial and mucosal tissues, these cells act as local stress sentinels that can present cytotoxic, IL-17A-producing, or regulatory programs depending on microenvironmental cues.^{30 31} They also undergo antigen-driven clonal expansion, supporting long-term tissue surveillance.^{32–34}

Murine $\gamma\delta$ T cells: innate and adaptive attributes

In mice, the balance between innate-like and adaptive-like behavior is determined by thymic programming.¹⁴ Fetal-derived $V\gamma 5^+$ sense keratinocyte stress (including NKG2D ligands) and rapidly release IFN- γ , IGF-1, and KGF to foster wound healing and barrier integrity.^{15 18}

$\gamma\delta^{17}$ -biased subsets such as $V\gamma 6^+$ and part of $V\gamma 4^+$ cells adopt strongly innate-like profiles, tissue residency, and cytokine-driven activation with minimal peripheral antigen reshaping.^{1 24} Their activation is shaped by inflammatory cytokines (IL-1 β , IL-23, and IL-7), stress-induced ligands (eg, NKG2D ligands such as Rae-1, Mult1, and H60), as well as commensal-derived signals, allowing for rapid responses in the absence of (previous) antigen-driven expansion to provide immediate protection at frontline surfaces.^{8 38 40 41} These cells rapidly express IL-17A, amphiregulin, and IGF-1, as well as other mediators that are important components of epithelial homeostasis and tissue repair, in addition to barrier immunity, promoting the recruitment of neutrophils and macrophages.¹⁹ Beyond TCR-mediated sensing, murine $\gamma\delta$ populations also integrate PRR signals (toll-like receptor (TLR), nucleotide-binding oligomerization domain (NOD)), which can directly activate $\gamma\delta$ T cells in

highly inflammatory settings and amplify cytokine-driven effector responses.⁹

On the other hand, $\gamma\delta^{\text{IFN}\gamma}$ cells such as $V\gamma 1^+$ and some $V\gamma 4^+$ populations exhibit more adaptive-like features.^{8 42} These populations have diverse TCR repertoires, undergo antigen-dependent expansion following infection or inflammation, and can generate memory-like recall responses.²⁵ Cytokines such as IL-12 and type I IFNs, together with co-stimulatory signals, promote T-bet/Eomes-driven cytotoxic programs.⁴⁰ Under strong activation, murine $\gamma\delta^{\text{IFN}\gamma}$ cells may transiently acquire antigen-presenting cell-like features, including expression of MHC-II and co-stimulatory molecules, enabling short-lived antigen presentation to $\alpha\beta$ T cells in inflammatory contexts.^{13 30 31} Thus, antigen presentation is not a dominant physiological role in mice but can emerge in specific inflammatory settings.^{24 38 43}

Human $\gamma\delta$ T cells: balancing innate reactivity and adaptive plasticity

Human $\gamma\delta$ T cells are shaped post-thymically by diverse environmental stimuli, chronic antigenic challenges, and local cytokine networks.^{38 44}

In the intestinal epithelium, BTN/BTNL family members impose tissue-specific, TCR-dependent tuning of $\gamma\delta$ IELs, representing an innate-like mode of immune instruction.²³ In humans, BTNL3–BTNL8 complexes expressed by intestinal epithelial cells preferentially engage and regulate $V\delta 2^-$ $\gamma\delta$ IELs.^{23 34} Human $V\delta 2^-$ $\gamma\delta$ IELs display residency programs and IL-7R expression that are consistent with long-term epithelial maintenance rather than classical adaptive priming.^{23 42}

Focusing on $V\delta 2^+$ T cells, $V\gamma 9^+$ and $V\gamma 9^-$ populations display distinct biology.²⁷ The $V\gamma 9^+$ subset has an innate-like phenotype relying on rapid phosphoantigen sensing through BTN3A1/BTN2A1,^{6 45} and PRRs further amplify IFN- γ production and cytotoxicity through MyD88–mTORC1–NF- κ B/MAPK signaling.^{29 46} They produce chemokines and recruit NK cells, monocytes, neutrophils, dendritic cells, and other immune effectors.⁴⁴ Under strong stimulation, these cells can acquire transient APC-like properties and reversible regulatory functions, suppressing $\alpha\beta$ T-cell proliferation via programmed death-ligand 1 (PD-L1)/cytotoxic T-lymphocyte associated protein 4 (CTLA-4) pathways and cytokine release (TGF- β 1, IL-15).^{26 39 44} Despite their innate bias, under strong microbial or metabolic stimulation, selected $V\gamma 9V\delta 2^+$ clonotypes can undergo clonal expansion and differentiation, conferring a quasi-adaptive layer onto their effector responses.²⁷ In contrast, $V\gamma 9^- V\delta 2^+$ cells do not rely on BTN3A1/BTN2A1 and undergo antigen-driven clonal selection during infection or tissue stress.²⁷ In acute cytomegalovirus (CMV) infection, they can expand and differentiate into long-lived cytotoxic effectors, including in liver tissue, consistent with a role in solid organ immunosurveillance.^{27 47}

At epithelial and mucosal sites, $V\delta 1^+$ cells act as local sentinels recognizing stress-induced ligands through

natural killer group 2, member D(NKG2D)/DNAX accessory molecule-1 (DNAM-1), and in defined contexts can also respond to CD1-presented lipids or endothelial protein C receptor (EPCR)-associated signals, enabling detection of epithelial stress or transformation independently of phosphoantigens.^{33 35} Their effector polarization is shaped by local cytokines (IL-15, IL-1 β , IL-23, TGF- β), metabolic conditions (hypoxia, lactate), and transcription factor (TF) modules (T-bet, PLZF, ROR γ t, Eomes), supporting cytotoxic, IL-17A, or tissue-repair programs together with antigen-driven clonal expansion and long-term persistence.^{33 44}

$V\delta 1^+$ and $V\delta 3^+$ cells undergo oligoclonal expansion in chronic viral infection, epithelial stress, or sustained antigen exposure, reflecting adaptive remodeling.³³ $V\delta 3^+$ T cells can also respond to lipids and express NK-associated receptors, enabling detection of viral or metabolic stress.³⁶ Moreover, these cells have been reported to even interact with B cells, inducing maturation and isotype/class switching.⁹ $V\delta 5^+$ cells follow similar principles of stress-ligand recognition and context-dependent differentiation, with potential roles in early epithelial surveillance.⁹ $V\delta 3^+$ and $V\delta 5^+$ may play specialized roles in antiviral defense and hepatocellular carcinoma.^{35 48}

$\gamma\delta$ T CELLS IN TUMOR SURVEILLANCE

$\gamma\delta$ T cells play a role in tumor surveillance alongside conventional $\alpha\beta$ T cells and other immune cells by recognizing different stimuli independently of classical MHC.² They sense metabolic dysregulation, DNA damage, hypoxia, and epithelial stress via different receptors and pathways.^{1 3 5 49} $\gamma\delta$ TCR along with these receptors supports broad stress surveillance, while tissue-resident phenotypes enable rapid interception of malignant transformation.^{8 28}

Functionally, IFN- γ -producing $\gamma\delta$ T cells have direct cytotoxicity, enhance antigen presentation, polarize myeloid cells, recruit M1/cDC1, and license DCs for effective $\alpha\beta$ T-cell priming.² Furthermore, pan-cancer analyses consistently identify $\gamma\delta$ T-cell signatures with favorable survival, including in low-mutational burden tumors.⁵⁰

In contrast, IL-17A-producing $\gamma\delta$ T cells can promote myeloid, supporting angiogenesis and, under chronic inflammation, facilitating tumor progression.^{12 51 52} Thus, $\gamma\delta$ are protective or pathogenic depending on subset composition, transcriptional wiring, and the metabolic and cytokine landscape of the TME.¹ Immune checkpoints (PD-1, CTLA-4, NKG2A, killer-cell immunoglobulin-like receptor (KIRs)) further tune $\gamma\delta$ effector activity under chronic stimulation.^{2 47 53 54}

Mouse $\gamma\delta$ T cells in tumor surveillance

Within murine $\gamma\delta$ T cells, the contribution to protumor or antitumor immunity varies depending on the subset.^{51 55} In carcinogen-induced and spontaneous models, TCR $\delta^-/-$ exhibit higher incidence and faster progression of skin, mucosal, and soft-tissue tumors,

establishing $\gamma\delta$ T cells as a non-redundant barrier against malignant transformation.⁵⁵

The $V\gamma 5^+$ population can detect keratinocyte stress following ultraviolet irradiation, mutagen exposure, or oncogene activation and rapidly produce IFN- γ /TNF- α and repair-associated mediators, coupling tumor control to epithelial repair.¹⁵

$V\gamma 6^+$ cells and $\gamma\delta^{17}$ $V\gamma 4^+$ cells can promote tumor progression through induction of C-X-C motif chemokine ligand 5 (CXCL5)/granulocyte colony-stimulating factor (G-CSF), recruitment of myeloid-derived suppressor cells (MDSCs) and neutrophils, angiogenesis, and supporting early metastatic seeding.^{19 20} In colorectal cancer models, these types of cells accumulate within tumor lesions, whereas adjacent tissue is enriched for $\gamma\delta^{IFN-\gamma}$ cells, highlighting a spatial segregation of protumor versus anti-tumor $\gamma\delta$ responses.^{19 51}

Conversely, $V\gamma 1^+$ and $\gamma\delta^{IFN-\gamma}$ $V\gamma 4^+$ cells mediate anti-tumor responses via IFN- γ and cytotoxic pathways (perforin/granzyme B, TNF-related apoptosis-inducing ligand (TRAIL), Fas ligand (FasL)) and can engage TCR-dependent, NK-receptor-dependent and Fc γ RIII-dependent mechanisms.⁴¹ In models of lymphoma, prostate carcinoma, breast cancer, and lung metastasis, the depletion or genetic dysfunction of these $\gamma\delta^{IFN-\gamma}$ subsets accelerates tumor growth and dissemination.⁵¹ Their IFN- γ enhances MHC expression on tumor/APCs, supports M1 polarization, promotes antibody-dependent cellular cytotoxicity (ADCC)-like effects, and favors cDC1-driven cross-priming.^{19 35 56}

Human $\gamma\delta$ T cells in tumor surveillance

In humans, tumor surveillance is dominated by circulating $V\delta 2^+$ cells and tissue-resident $V\delta 2^-$ subsets, shaped by cytokines, phosphoantigens, stress-ligands, and microbial pathogen-associated molecular patterns (PAMPs).^{29 38 43}

$V\gamma 9V\delta 2^+$ cells sense dysregulated mevalonate-pathway metabolites (Isopentenyl diphosphate (IPP), (E)-4-Hydroxy-3-methyl-but-2-enyl diphosphate (HMBPP)), rapidly deploy cytotoxicity, and can induce immunogenic cell death, expose damage-associated molecular patterns (DAMPs), and enhance M1/DC recruitment and maturation.⁵⁷ AMPK-driven metabolic stress increases BTN2A1/BTN3A1 surface expression, thereby sensitizing tumor cells to $V\gamma 9V\delta 2^+$ recognition and enhancing $\gamma\delta$ T-cell surveillance.⁵⁸ On strong activation, $V\gamma 9V\delta 2^+$ cells upregulate MHC-II, CD80/CD86, and CD40, cross-present antigens to CD4 $^+$ and CD8 $^+$ T cells.⁴⁶ Moreover, they can co-stimulate NK and CD8 $^+$ T cells via 4-1BBL, amplifying antitumor immunity. (4) (49) Clinically, prescription drugs used to prevent skeletal-related events in metastatic disease (eg, pamidronate or zoledronate) induce intracellular phosphoantigen accumulation, thereby supporting $V\gamma 9V\delta 2^+$ T-cell activation through the BTN3A1–BTN2A1 metabolic surveillance axis.^{32 52 59} A recent analysis of nearly 40 patients with glioblastoma demonstrated that autologous tumors were recognized by $\gamma\delta$ T cells, but not $\alpha\beta$ T cells.⁶⁰ Furthermore, $V\gamma 2^+$ $\gamma\delta$ tumor-infiltrating

lymphocytes (TILs) provided consistently effective responses following phosphoantigen sensitization, and $V\gamma 2^-$ $\gamma\delta$ responses occurred in limited patients.⁶⁰ Although higher $V\delta 2^+$ $\gamma\delta$ TIL frequencies were associated with an increased overall survival, cytotoxic activity of these cells varied substantially among patients.⁶⁰ This heterogeneity of responses was due to immune-intrinsic and tumor-intrinsic characteristics, including the availability of BTN2A1/BTN3A complexes, as well as extracellular matrix and TGF- β -associated programs which limit the fitness of $\gamma\delta$ T cells.⁶⁰ Together, these findings illustrate the importance of tumor architecture and ligand accessibility, which can shape $\gamma\delta$ responsiveness and explain clinical variability.

In most solid tumors, however, the dominant $\gamma\delta$ TILs are $V\delta 2^-$, particularly $V\delta 1^+$ and $V\delta 3^+$ cells with tissue-resident memory (TRM)-like phenotypes.³¹ These recognize stress ligands (MICA/B, ULBPs), CD1-presented lipids, EPCR, EphA2, and annexin A2 and frequently show private clonal expansions in kidney, colorectal, lung, pancreatic, and hepatobiliary cancers.^{2 3 48} $V\delta 1^+/V\delta 3^+$ TILs display cytotoxic TRM-like programs, and their abundance often correlates with favorable prognosis.^{33 36 53} Notably, this effector function depends on the immunosuppressive pressure of the TME.⁶¹ As seen in microsatellite-stable colorectal cancer, $V\delta 1^+$ TILs can become dysfunctional because of inflammatory fibroblasts suppressing their activity via the T cell Immunoreceptor with Ig and ITIM domains (TIGIT)–Nectin cell adhesion molecule 2 (NECTIN2) axis, with TIGIT blockade partially restoring cytotoxic function.⁶¹ In contrast, in renal cell carcinoma (RCC) and urothelial cancer, $V\delta 2^-$ $\gamma\delta$ T cells in RCC often acquire a transcriptional program resembling terminal $\alpha\beta$ T-cell exhaustion (PD-1 $^+$ TIGIT $^+$ T cell immunoglobulin and mucin-domain containing protein 3 (TIM-3 $^+$)), retain perforin/granzyme expression, and exhibit clonal expansion and potent cytotoxicity.⁵³ Their $V\delta 2^-$ gene signature strongly predicted clinical benefit to anti-PD-1/PD-L1 therapy in RCC and urothelial cancer, indicating that exhausted-appearing $\gamma\delta$ T cells can remain dominant effectors in vivo.⁵³ In mismatch repair (MMR)-deficient colorectal cancer with B2M loss and HLA-I deficiency, PD-1 $^+$ $V\delta 2^-$ $\gamma\delta$ T cells can expand and become key mediators of checkpoint responses.^{2 35 41 62} Next-generation repertoire sequencing has shown that $V\delta 1^+$ cells can clonally focus from naïve CD27 $^+$ pools into long-lived CX3CR1 $^+$ cytotoxic effectors, supporting adaptive-like surveillance.³³

IL-17A-producing $\gamma\delta$ T cells can be protumorigenic in several epithelial cancers, particularly colorectal carcinoma, where high infiltration correlates with poor prognosis.⁵¹ They produce IL-17A, IL-22, IL-4, IL-10, TGF- β , and vascular endothelial growth factor (VEGF), recruiting and licensing MDSCs, M2 and neutrophils, promoting angiogenesis, and sustaining an immunosuppressive TME.⁶³ Similar accumulation is observed in gastric MALT lymphoma and other IL-1 β /IL-23-rich settings, where

IL-17A-phenotype cells track with enhanced myeloid infiltration.⁵² In breast cancer, IL-1 β -activated $\gamma\delta$ T cells drive a systemic IL-17A/G-CSF axis that expands suppressive neutrophils and fuels multiorgan metastasis, illustrating how $\gamma\delta$ T cells can orchestrate pro-metastatic inflammation.⁶⁴ In addition, V δ 1⁺ $\gamma\delta$ T cells can acquire regulatory phenotypes under the influence of TGF- β , IL-4, or tumor-derived signals.¹⁵ These cells express CD39, CD73, PD-1/PD-L1, or galectin-1, suppressing $\alpha\beta$ T cell and DC activity, promoting regulatory T cell (Treg) expansion through adenosine and, in their amphiregulin tissue-repair mode, may inadvertently support tumor growth during chronic inflammation.^{19 31 63} These regulatory programs are reinforced by IL-15/TGF- β signaling with PLZF, Egr2, and BLIMP.^{10 26} Moreover, rarer V δ 5⁺ cells produce IL-10, suppressing neutrophil recruitment and promoting wound healing, representing a repair-oriented arm that may become protumorigenic under chronic inflammation.^{15 36} Chronic microbiota-driven inflammation, hypoxia, lactate accumulation, obesity, aging, and persistent tumor-myeloid- $\gamma\delta$ crosstalk all favor these protumor $\gamma\delta$ and regulatory states.^{51 64}

At the mechanistic level, IL-12 promotes cytotoxic T-bet/Eomes programs, whereas IL-23/IL-1 β /TGF- β favors $\gamma\delta$ -IL-17A differentiation, and IL-15/TGF- β supports IL-10-producing regulatory states; together with stress-ligand engagement and metabolic stress, these cues determine whether $\gamma\delta$ T cells are antitumor or protumor in human tumors.^{51 65 66} Overall, the outcome depends less on $\gamma\delta$ abundance than on how the TME polarizes their effector states.

IMPLICATIONS FOR IMMUNOTHERAPY WITH $\gamma\delta$ T CELLS

Given their MHC-independent recognition, intrinsic stress sensing, and low alloreactivity, $\gamma\delta$ T cells can be used in immunotherapies when tumors have evaded $\alpha\beta$ T-cell surveillance.^{2 52}

Adoptive transfer and expansion approaches

Peripheral blood V γ 9V δ 2⁺ T cells are the most accessible subset for clinical-grade expansion.⁷ Phosphoantigens or aminobisphosphonates plus IL-2/IL-15 for maximal ex vivo expansion, and second-generation protocols providing feeder cells (such as K562-based artificial APCs) increase yield with minimization of activation-induced dysfunction.⁵⁴ Early clinical-scale studies demonstrated that K562-derived artificial antigen-presenting cells (aAPCs) expressing CD137L, membrane IL-15 and costimulatory ligands, together with IL-2 and IL-21, can drive >10³–10⁷-fold expansion of highly pure, polyclonal $\gamma\delta$ T cells while preserving V δ 1⁺, V δ 2⁺, and V δ 1-V δ 2⁺ diversity, supporting their feasibility for adoptive transfer.^{54 67} In early-phase trials, adoptive transfer of these cells has demonstrated antitumor activity against acute myeloid leukemia (AML), multiple myeloma, and lymphoma as well as several solid tumors, especially when combined with monoclonal antibodies or bisphosphonates.⁴

Importantly, the clinical feasibility and safety of allogeneic V γ 9V δ 2 T-cell therapy have now been demonstrated at scale.⁶⁸ In a phase I trial of 414 infusions given to 132 patients with advanced solid tumors, adoptive transfer was well tolerated, with no graft-versus-host disease (GvHD) or CRS.⁶⁸ In particular, for certain subsets of patients with liver and lung cancer who received multiple infusions (≥ 5 doses), survival was significantly extended, indicating preliminary efficacy.⁶⁸ Collectively, these data provide a key clinical milestone, in support of safety, feasibility and biological rationale for the use of off-the-shelf $\gamma\delta$ T-cell platforms. In relapsed/refractory AML, V γ 9V δ 2⁺ products such as TCB-202-001/TCB-008 induce transient remissions with excellent safety (no severe CRS, immune effector cell-associated neurotoxicity syndrome (ICANS), or GvHD), and donor-derived $\gamma\delta$ infusions after haplo-HSCT (eg, INB-100) appear feasible, although the durability of benefit remains to be defined.^{54 69 70}

By contrast, clinical-grade V δ 1⁺ expansion has been more challenging because of their low frequency in blood and phenotypic plasticity.^{7 9 39 71} Sequential IL-4 followed by IL-15 protocols enable generation of delta one T cells (DOT cells).⁷² DOT cells are highly polyclonal, NKR^{hi}, and have potent antileukemic activity while having minimal hematopoietic toxicity; these products are now in early-phase testing (eg, TK012/TK02) in relapsed/refractory AML.⁷² AML blasts express high levels of ULBP1/2 and MICA/B and secrete CXCL8/10 and CCL5, supporting chemokine-driven recruitment of both V γ 9V δ 2⁺ and V δ 1⁺ cells and providing a strong biological rationale for deploying both subsets in myeloid malignancies.^{54 70 72}

Recent work on tissue-resident V δ 1⁺ $\gamma\delta$ cells highlights their suitability for solid tumors.³⁹ These cells acquire TRM-like programs that support durable local surveillance, rapid recall responses, and recognition of epithelial stress ligands, positioning V δ 1⁺-based platforms as promising adoptive therapies for colorectal, hepatocellular, and pancreatic cancers.^{71 73}

Checkpoint blockade and combination therapies

Immune checkpoint blockade (ICB) developed for $\alpha\beta$ T cells also modulates $\gamma\delta$ T-cell responses.⁶²

Extensive patient studies have shown that ICB can dynamically remodel checkpoint receptor landscapes in $\gamma\delta$ T cells, with marked divergence between V δ 1 and V δ 2 subsets.⁷⁴ In melanoma, V δ 1⁺ cells progressively acquire the expression of inhibitory receptors (PD-1 and TIGIT) and TOX-associated and KIR-associated transcriptional modules, yet they retain substantial cytotoxic activity.⁷⁴ This highlights that the expression of inhibitory receptors is uncoupled from functional exhaustion.⁷⁴ Notably, binding of a therapeutic anti-PD-1 antibody to V δ 1⁺ $\gamma\delta$ T cells increases markedly in responders and correlates with clinical benefit, supporting a pharmacodynamic and possibly prognostic biomarker role.⁷⁴ Taken together, these findings suggest that $\gamma\delta$ T cells should not be treated as a uniform checkpoint target, but they support the use

of subset-informed stratification and rational combination or engineering strategies.

PD-1/PD-L1 or CTLA-4 blockade can restore $\gamma\delta$ cytotoxicity and cytokine production, while IL-12 and IL-15 support persistence.^{53 75} Across tumor types, $\gamma\delta$ T cells can retain effector function under chronic antigenic pressure and may outperform exhausted $\alpha\beta$ T cells in MHC-I-deficient environments.² In renal cell and urothelial carcinomas, V δ 2⁻ $\gamma\delta$ T cells (predominantly V δ 1⁺) frequently dominate responses to PD-1 blockade, retaining strong cytotoxic programs despite PD-1⁺TIGIT⁺TIM-3⁺ “exhausted-like” phenotypes and correlating with improved survival, thereby establishing $\gamma\delta$ immunity as a dominant effector axis when conventional $\alpha\beta$ T-cell immunity is compromised by HLA-I defects and/or profound exhaustion-associated transcriptional dysfunction.^{2 53 69 75 76}

Rational strategies, therefore, include combining ICB with $\gamma\delta$ adoptive transfer in MHC-low tumors, PD-1/TIGIT blockade to restore $\gamma\delta$ activity, and engineering checkpoint-resistant $\gamma\delta$ products (eg, PD-1-knockout CAR- $\gamma\delta$ cells) supported by IL-15 superagonists or PI3K modulation.^{7 62 77} An alternative approach uses engineered $\gamma\delta$ cells to secrete anti-PD-1 antibodies, which can locally neutralize PD-1/PD-L1 signaling and enhance cytotoxicity and persistence in solid tumors.⁷⁸

CAR $\gamma\delta$ engineering immunotherapies

$\gamma\delta$ T cells combine stress-sensing TCR with NK receptors, making them attractive CAR platforms.⁶⁹ As CAR- $\gamma\delta$ designs diversify, organoid-on-chip platforms are increasingly valuable to benchmark infiltration, serial killing, and resistance to TME suppression before moving into mouse models.⁷⁹

Preclinical studies showed that V γ 9V δ 2⁺ expressing CARs against CD19, GD2, or EGFR produce high IFN- γ and eliminate leukemia and neuroblastoma cells in vitro and in vivo.⁸⁰ Non-viral delivery using the Sleeping Beauty transposon system enabled efficient insertion of CD19 CARs into mixed V δ 1⁺/V δ 2⁺ populations, generating potent polyclonal CAR- $\gamma\delta$ products.³⁸ NKG2D-based CARs built on V γ 9V δ 2⁺ cells expand recognition of NKG2D-ligand-expressing solid tumors while preserving endogenous $\gamma\delta$ TCR-mediated stress-sensing.^{38 80}

Importantly, $\gamma\delta$ T cells retain CAR-independent killing pathways, including TRAIL-DR4/DR5 and Fas-FasL interactions, providing backup cytotoxicity when CAR antigen density is low and potentially limiting antigen escape.⁸⁰ IL-21-reprogrammed V δ 1⁺ cells can now be expanded as metabolically fit cytotoxic products that support precise AAVS1-targeted CAR insertion and show strong activity in solid-tumor models, positioning V δ 1⁺ CARs as off-the-shelf candidates for epithelial cancers.⁸¹

Clinically, ADI-001 (allogeneic V δ 1⁺ anti-CD20 CAR- $\gamma\delta$) showed high response rates in heavily pretreated lymphoma, including post-CAR- $\alpha\beta$ failures, with an excellent safety profile.^{54 70} For solid tumors, B7-H3 CAR-V γ 9V δ 2⁺ and CAR-V δ 1⁺ cells showed potent preclinical

activity, and the allogeneic V δ 1⁺ B7-H3 CAR UTAA06 is advancing toward clinical testing.⁶⁶ Additional constructs, including epidermal growth factor receptor (EGFR)-CARs and PD-L1-CARs, are being developed for antigen-heterogeneous tumors.^{82 83} Head-to-head comparisons indicate that CAR-V δ 1⁺ cells can match CAR- $\alpha\beta$ efficacy with lower toxicity and no detectable GvHD or severe CRS, while single-cell analyses reveal reduced exhaustion and greater proliferative resilience, supporting prioritization of V δ 1⁺ CAR platforms for allogeneic use.^{82 83}

Beyond genetic methods, non-viral chemical surface engineering can rapidly retarget $\gamma\delta$ cells. (78) (85) Metabolic labeling of cell-surface sialic acid with AMS-ManNAz-P followed by click chemistry now makes dense conjugation of tumor-specific antibodies possible without disruption to BTN3A1/BTN2A1, NKG2D, or DNAM-1 signaling.^{77 84} These antibody- $\gamma\delta$ conjugates enhance killing, induce Gasdermin3 (GSDME)-dependent pyroptosis with ATP/High-mobility group box 1 (HMGB1) release, and remodel the TME via C-C Chemokine Receptor (CCR5)/C-C motif chemokine ligand 5 (CCL5)-dependent CD8⁺ T-cell recruitment.^{77 84} In ovarian-cancer xenografts, α PD-L1- $\gamma\delta$ conjugates improved tumor control and survival compared with unmodified $\gamma\delta$ cells.⁸⁴

In addition to stable viral or transposon-based editing, transient messenger RNA electroporation can also be used to express CARs or bispecific T-cell engagers, enhancing $\gamma\delta$ cytotoxicity against B-cell acute lymphoblastic leukemia (B-ALL) while preserving endogenous stress-sensing and potentially reducing long-term toxicities.⁸⁵ In bone-metastatic castration-resistant prostate cancer, $\gamma\delta$ -enriched prostate stem cell antigen (PSCA) CAR-T cells induced rapid regression of intratibial tumors and, particularly in combination with zoledronate, enhanced antitumor efficacy and protection from cancer-induced bone disease while preserving V γ 9V δ 2⁺ TCR function.⁵⁹

Donor selection is emerging as another lever to optimize these strategies.⁶⁹ Selection of CD16^{hi} V γ 9V δ 2⁺ cells displays superior cytotoxicity and ADCC, with reduced Th17 polarization.⁸⁶ Engineered with CAR/IL-15, these cells show durable persistence, dual CAR/TCR targeting, and strong activity in ovarian-cancer xenografts without GvHD.⁸⁶

Payload-delivering $\gamma\delta$ T Cells

Beyond CARs, a complementary strategy uses V γ 9V δ 2⁺ T cells as secretory effector hubs.⁸⁷ Fowler *et al* engineered V γ 9V δ 2⁺ cells to secrete a tumor-targeting single-chain variable fragment-Fc fusion (scFv-Fc) opsonin together with an IL-15 α -IL-15 fusion (stIL-15-OPS- $\gamma\delta$).⁸⁷ These cells demonstrate augmented antigen-specific cytotoxicity, providing IL-15-dependent persistence in the absence of added cytokines.⁸⁷ Moreover, they recruit NK cells, macrophages, and neutrophils in addition to host $\gamma\delta$ T cells through Fc γ R and IL-15 signaling.⁸⁷ In patient-derived osteosarcoma models, they had better outcomes than CAR- $\alpha\beta$ T cells, especially with zoledronic acid,

highlighting $\gamma\delta$ T cells as payload-delivering immune activators in bone-resident and therapy-resistant tumors.⁸⁷

TCR-engineered $\gamma\delta$ and $\gamma\delta$ -inspired $\alpha\beta$ T cells

A similar axis of innovation seeks to use $\gamma\delta$ TCRs as modular therapies, instead of transferring $\gamma\delta$ cells.⁷⁸⁸ TCR-engineered $\gamma\delta$ (TEG) approaches convert tumor-reactive V γ 9V δ 2⁺ TCRs into $\alpha\beta$ T cells, which recognize a wide range of stress ligands while only inducing limited allo-reactivity.⁸⁹ TEG001 and related agents have promising outcomes in AML xenograft models and are advancing towards early clinical testing.⁷⁸⁸ AbTCR/ARTEMIS receptors, which fuse an antibody Fab to $\gamma\delta$ -TCR signaling chains, preserve MHC-independent recognition, prevent TCR mispairing, and reduce tonic signaling.^{88 90} Early trials in lymphoma and hepatocellular carcinoma report durable responses.^{88 90}

Bispecific engagers and $\gamma\delta$ -activating antibodies

$\gamma\delta$ bispecific and agonist antibodies offer cell-free strategies to activate endogenous $\gamma\delta$ T cells in situ.³² V γ 9-targeting bispecifics against HER2, CD40, CD1d, CD123, or CD19 enhance V γ 9V δ 2⁺-mediated cytotoxicity, ADCC, and phagocytosis.^{32 69 89} $\gamma\delta$ TCR anti-CD3 bispecific molecules (Gamma delta TCR anti-CD3 bispecific molecules (GABs)), built from high-affinity V γ 9V δ 2⁺ TCRs, confer BTN2A1/BTN3A1-dependent and phosphoantigen-dependent tumor recognition to bulk $\alpha\beta$ T cells, enabling killing of AML and myeloma cells in three-dimensional (3D) bone-marrow models, while sparing healthy hematopoietic and stromal cells.⁹¹ Clinical agents, such as ICT01 (BTN3A agonist) and LAVA-051 (CD1d $\times$ V δ 2⁺ bispecific), activate V γ 9V δ 2⁺ and invariant NK cells with limited CRS and are being evaluated in AML in combination with hypomethylating agents and venetoclax.^{32 92} $\gamma\delta$ engagers provide an off-the-shelf means of $\gamma\delta$ activation in tumors with impaired antigen presentation.

Future directions and challenges

$\gamma\delta$ T cells have strong therapeutic potential, but several barriers still limit broad clinical translation.⁶⁹ Key challenges include phenotypic plasticity, donor-dependent heterogeneity, scalable and reproducible manufacturing, and control of off-target activity.⁷²

In myeloid malignancies, the required immune evasion by the tumor further limits efficacy: malignant blasts use PD-L1, galectin-9, and CD112/CD155 to engage PD-1, TIM-3, and TIGIT on $\gamma\delta$ T cells, while arginase-producing MDSCs suppress V γ 9V δ 2⁺-derived IFN- γ , emphasizing the need to reverse these suppressive factors for durable responses.⁵⁴⁹³ Manufacturing should therefore supplement $\gamma\delta$ cytotoxic subsets with an IFN- γ bias while constraining and regulating protumoral phenotypes.⁹³ Beyond cellular engineering, optimized immunostimulatory agents, including PRR agonists, protein adjuvants, bispecific molecules, and virus-like particle-based platforms, may enable in vivo $\gamma\delta$ T-cell activation and strengthen antitumor responses in situ.^{5 32 46 94 95} Integration with single-cell and

TCR-repertoire analytics will be essential to sharpen specificity and safety in next-generation $\gamma\delta$ therapies.^{96 97}

EXPERIMENTAL PLATFORMS FOR STUDYING $\gamma\delta$ T CELLS IN CANCER

The renewed interest in $\gamma\delta$ T cells in oncology reflects their unique biology and application versatility.⁹⁸ A variety of experimental platforms, from reductionist co-cultures and 3D tumor organoids to genetically engineered mouse models and humanized xenografts enable mechanistic dissection of $\gamma\delta$ T-cell function under increasingly physiologic TME conditions.⁹⁹

Organoid-immune co-cultures: a precision modeling breakthrough

Patient-derived organoids preserve tumor heterogeneity, 3D architecture, and genomic landscape. (100) (101) Co-cultures of $\gamma\delta$ T cells, CAR- $\gamma\delta$ cells, or TEGs with 3D tumor models (PDOs or spheroids) provide human-relevant readouts of cytotoxicity, serial killing, cytokine release, and dependency on specific stress-ligand pathways.⁷³

Recent advances in the field of organoid engineering have resulted in more standardized and physiologically relevant systems for assessing $\gamma\delta$ -based immunotherapy.⁷⁹ Zhu *et al* developed an integrated microgravity-microfluidic platform that preserves TME components, reconstitutes parental cell populations and mimics physiological conditions.⁷⁹ In this platform, they tested a CAR- $\gamma\delta$ T-cell therapy against glioblastoma organoids, investigating infiltration, serial killing, and combination regimens under defined TME-like constraints.⁷⁹ The use of such systems bridges the gap between in vitro and in vivo models.

High-dimensional systems such as BEHAV3D combine 3D live imaging with single-cell transcriptomics, to classify engineered T-cell behaviors (productive synapses, serial killing, and non-productive engagement) and link these to transcriptional states.^{2 30 100} Originally developed for CAR- $\alpha\beta$ T cells, BEHAV3D is now being adjusted for $\gamma\delta$ systems to assess motility, cell death, and resistance to TME suppression in CRC, RCC, glioma, and other PDO models.^{73 101 102} CAR- $\gamma\delta$ cells specific for EGFR, PD-L1, or folate receptor- α mediate potent organoid killing is enhanced when BTN3A1/BTN2A1 phosphoantigen sensing is maintained, highlighting the cooperation between CAR specificity and endogenous $\gamma\delta$ TCR recognition.⁷³

There are also now more standardized protocols for the type of matrices, cytokine cocktails, and imaging used to allow reproducibility across laboratories.^{46 71 73 103} As $\gamma\delta$ systems differ markedly between mice and humans, PDO-based platforms are essential for translating murine insights into human-relevant strategies.^{64 76}

Xenograft and humanized mouse models: in vivo validation of human $\gamma\delta$ therapies

Immunodeficient mice xenografted with human tumors and infused with human $\gamma\delta$ T cells are invaluable

models for preclinical testing to determine cytotoxicity, persistence, and migratory behavior.¹⁰⁴ NSG and NSG-SGM3 mice engrafted with human tumors and treated with $\gamma\delta$ T-cell products (including unmodified $\gamma\delta$ cells, CAR- $\gamma\delta$, DOT cells, TEGs, or stIL-15-OPS- $\gamma\delta$) enable analysis of in vivo toxicity, persistence, trafficking, and bystander effects.¹⁰⁴ stIL-15-OPS- $\gamma\delta$ cells demonstrate potent control of osteosarcoma xenografts with durable persistence and minimal toxicity.^{87 88} CAR- $\gamma\delta$ cells can eliminate glioblastoma xenografts even under complete MHC-I loss, underscoring their potential for antigen-presentation-deficient tumors.^{73 80}

Humanized mice reconstituted with human hematopoiesis allow study of $\gamma\delta$ interactions with DCs, NK cells, Tregs, and MDSCs.^{17 57} This is critical in myeloid malignancies where TGF- β , IDO, adenosine, and arginase-mediated pathways dominate immune suppression.^{17 57}

Because mice lack functional BTN3A1/BTN2A1 orthologs and display distinct $\gamma\delta$ TCR repertoire structures, integrated readouts across PDOs, BEHAV3D, and xenografts provide the strongest translational workflow.⁷³

High-dimensional profiling and spatial mapping

Single-cell RNA sequencing, cytometry by time-of-flight, multiplex immunohistochemistry, and spatial transcriptomics are redefining $\gamma\delta$ T-cell diversity in cancer.⁹⁷ These technologies identify clonally expanded, IFN- γ -biased $\gamma\delta$ populations in colorectal, hepatic, and pulmonary tumors associated with favorable prognosis, whereas IL-17A-dominated clusters correlate with tumor-promoting inflammation and immunosuppression.⁹⁶ Spatial mapping resolves the localization of $\gamma\delta$ T cells within tumor regions, including invasive margins and tertiary lymphoid structures, identifying regions where $\gamma\delta$ T cells interact with critical immune partners.^{34 50}

Integrated single-cell, spatial, and TCR-sequencing analyses in non-small cell lung cancer identified a dominant tissue-resident V δ 1⁺ compartment, with TRDV1 expression predicting relapse-free survival and PD-1-blockade benefit, illustrating how high-dimensional mapping reveals prognostic $\gamma\delta$ architectures missed by bulk assays.^{31 96} Advances in $\gamma\delta$ TCR-sequencing enable systematic discovery of high-affinity, tumor-reactive $\gamma\delta$ TCRs, accelerating development of TEG-like $\gamma\delta$ TCR-redirected therapies.^{4 31 33}

Together, these high-dimensional approaches provide a blueprint for $\gamma\delta$ -directed immunotherapies that exploit tissue-specific and niche-specific activation states rather than treating $\gamma\delta$ T cells as a uniform population.^{20 30 97}

THERAPEUTIC POTENTIAL AND CLINICAL CHALLENGES OF $\gamma\delta$ T IMMUNOTHERAPIES

$\gamma\delta$ T cells have emerged as versatile effectors for cancer immunotherapy, owing to unique immunological features making them well suited for therapeutic applications.⁹⁸ Despite this, overcoming cellular heterogeneity,

manufacturing scalability, and TME-driven immune evasion remains a major challenge.^{69 99}

Persistence and exhaustion

$\gamma\delta$ T cells can undergo terminal differentiation, metabolic dysfunction, and checkpoint-driven exhaustion in solid tumors or leukemic niches. (4) (55) (78) Nonetheless, the expression of exhaustion markers does not translate into loss-of-function as mentioned before, since V δ 1 TIL $\gamma\delta$ cells in humans may have extensive cytolytic activity despite the presence of exhaustion-like signatures, underscoring the need to distinguish antigen engagement from transcriptional dysfunction.⁷⁴

Similar to CAR- $\alpha\beta$ T-cell studies, these states are shaped by glucose restriction with lactate accumulation, arginine/tryptophan depletion, and hypoxia-induced adenosine via CD39/CD73.⁷⁷ Strategies to enhance persistence therefore focus on improving metabolic fitness and enforce mitochondrial biogenesis.^{2 77} Genome editing of inhibitory pathways (PD-1, TIGIT, TGF- β signaling) may further confer resistance to exhaustion, while epigenetic regulators are likely to influence $\gamma\delta$ persistence and may become future engineering levers.⁷⁷

Tumor trafficking and TME immunosuppression

$\gamma\delta$ infiltration into solid tumors and bone niches is often suboptimal.⁵⁶ Chemokine receptor engineering and bisphosphonate co-administration improve migration, particularly in osteosarcoma and prostate bone metastases.^{9 59} Once within the TME, $\gamma\delta$ function is suppressed and can skew subsets toward IL-17A or regulatory phenotypes.^{7 73} Countermeasures include checkpoint blockade, dominant-negative TGF- β receptors, PD-1-resistant designs, and targeted ROR γ t suppression to preserve IFN- γ -biased programs.⁹³

Manufacturing and subset choice

Finally, manufacturing and subset choice remain complex.^{52 59 91} V γ 9V δ 2⁺ cells are readily expandable and well suited for allogeneic use, whereas V δ 2⁻ subsets, despite exhibiting greater donor variability and plasticity, dominate solid-tumor infiltrates and display superior tissue-resident behavior.^{60 73}

This has driven parallel V δ 2⁺-centered and V δ 2⁻-centered pipelines within the same indications.⁶²

Next-generation strategies

The $\gamma\delta$ field is moving towards modular, personalized, and integrative designs.⁶⁹ Smart CAR- $\gamma\delta$ platforms incorporate logic gating, suicide switches, IL-15 armoring, metabolic optimization to sustain function within the TME, and payload-delivery strategies.¹⁰² PDO-guided personalization will enable patient-specific selection of targets, $\gamma\delta$ subsets and rational combinations.¹⁰⁰ In parallel, antigen-specific in situ stimulation strategies may enable selective in vivo expansion of defined $\gamma\delta$ subsets without adoptive transfer, providing a complementary, cell-free therapeutic route.^{4 100} Integrated $\gamma\delta$ -based strategies will likely combine $\gamma\delta$ products with $\alpha\beta$ CAR-T, NK cells, DC

vaccines, ICB, and standard regimens (eg, azacitidine/venetoclax).³⁸ Ultimately, translating $\gamma\delta$ biology into durable patient benefit will depend on scalable manufacturing, sustained persistence, effective trafficking, and avoidance of immunosuppression within the TME.⁹⁹

CONCLUSION AND FUTURE DIRECTIONS

$\gamma\delta$ T cells stand at the boundary between innate and adaptive immunity. The capability to rapidly detect cellular stress independent of MHC molecules remains a unique functional flexibility. In this review, we have outlined the major features of $\gamma\delta$ T-cell biology, including their development, variety in distinct species, and role in tumor surveillance. A crucial aspect is the extensive functional plasticity of $\gamma\delta$ T cells, which can acquire protumoral or antitumoral phenotypes depending on the environmental stimuli. Therefore, for clinical application of $\gamma\delta$ T cells, the choice of subset, activation status and disease context should be carefully considered. Progress in experimental systems has informed our views on $\gamma\delta$ T-cell behavior, and new platforms such as microfluidic and organ-on-chip technologies have the potential to expand the available tools. From a therapeutic perspective, $\gamma\delta$ -based strategies enable unprecedented opportunities for off-the-shelf immunotherapies, particularly in tumors resistant to conventional $\alpha\beta$ T-cell immunity. Despite the challenges to be addressed, $\gamma\delta$ T cells could revolutionize cancer immunotherapy by expanding the rules of immune recognition and response beyond the limitations of conventional T-cell paradigms.

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